Stepper motor

DO | RASPBERRY PI | OUTPUT | MOVEMENT

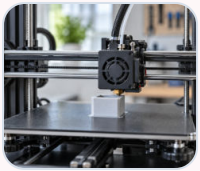
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

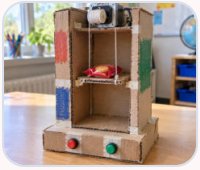
The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



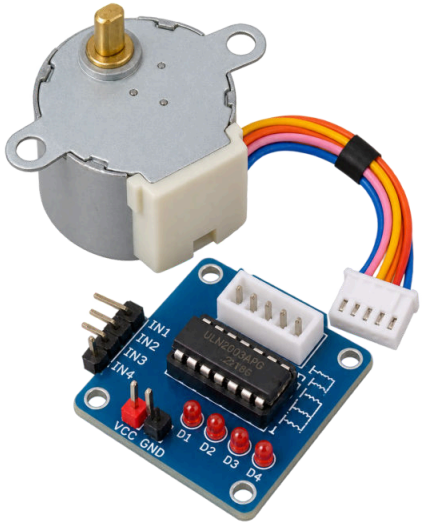
3D printer positioning

DETECT Read next print position → **DECIDE** How many steps are needed? → **DO** Move the print head



Snack lift

DETECT Snack request arrives → **DECIDE** Which floor needs it? → **DO** Step lift into place



Output shaft Moves in measured steps

Five-wire plug Connects to the driver

IN1 to IN4 Coil sequence

5V and GND Driver power

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

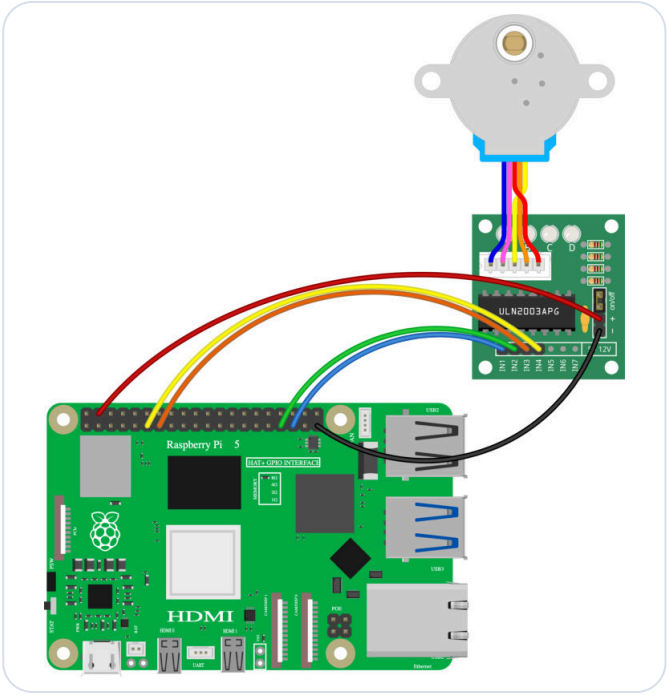
coils = [OutputDevice(pin) for pin in (19, 13, 27, 17)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence * 512:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

`sequence * 512` runs about one full turn. Reverse the sequence to turn the other way.

Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **19, 13, 27, 17** (pins 35, 33, 13, 11); driver → **5V** (pin 4) and **GND** (pin 39)